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B.Tech / M.Tech (Integrated) DEGREE EXAMINATION, NOVEMBER 2024

Fourth Semester

21CSC204J - DESIGN AND ANALYSIS OF ALGORITHMS

(For the candidates admitted from the academic year 2021-2022 to 2023-2024)

Note:

(i) **Part - A** should be answered in OMR sheet within first 40 minutes and OMR sheet should be handed over to hall invigilator at the end of 40th minute.

(ii) **Part - B & Part - C** should be answered in answer booklet.

Time: 3 hours

Max. Marks: 75

PART - A (20 x 1 = 20 Marks)

Answer ALL Questions

Marks BL CO PO

- Assume that $f(n)$ and $g(n)$ are asymptotically positive, then which of the following relation holds good? 1 1 1 1
 A) $f(n)=O(g(n))$ and $g(n) = O(h(n))$ then $f(n)=O(h(n))$ B) $f(n)=\Omega(g(n))$ and $g(n) = O(h(n))$ then $f(n)=\Omega(h(n))$
 C) $f(n)=\Omega(g(n))$ and $g(n) = \Theta(h(n))$ then $f(n)=\Omega(h(n))$ D) $f(n)=O(g(n))$ and $g(n) = \Theta(h(n))$ then $f(n)=\Omega(h(n))$
- The number of comparisons required to search $x=18$ in the list $A = [5,44,89,22,18,9,3,15,8]$ using Linear search 1 2 1 1
 A) 3 B) 1
 C) 4 D) 5
- What is the time complexity of the following code. Assume that $n>0$? 1 4 1 2

```
int segment(int n) {
    if(n==1)
        return 1;
    else
        return (n+ segment(n-1));
}
```

 A) $O(n)$ B) $O(\log n)$
 C) $O(n^2)$ D) $O(n!)$
- Which of the following functions provides the maximum asymptotic complexity? 1 2 1 2
 A) $f_1(n) = n^{(3/2)}$ B) $f_2(n) = n^{(\log n)}$
 C) $f_3(n) = n \log n$ D) $f_4(n) = 2^n$
- If for an algorithm, the time complexity is given by $O(\log^2 100)$ then upper bounded complexity will be ----- 1 1 2 1
 A) Polynomial B) Constant
 C) Linear D) Hyperbolic
- What is the optimal time required for solving the closest pair problem using divide and conquer approach? 1 1 2 1
 A) $O(N)$ B) $O(\log N)$
 C) $O(N \log N)$ D) $O(N^2)$
- The ----- is a set of points in the plane forms the smallest convex polygon for which each point is either on the boundary or in the interior of the polygon. 1 1 2 1
 A) Vertices B) Convex Hull
 C) Collinear D) Bridge

8. Which of the following sorting algorithms provide the best time complexity in the worst - case scenario? 1 3 2 2
 A) Merge Sort B) Quick Sort
 C) Bubble Sort D) Selection Sort
9. A city planner needs to connect all new residential areas with roads in such a way that the total length of the roads are minimized, but every area can be reached from any other. Which of the graph theory concept is more suitable for this kind of road planning approach? 1 1 3 2
 A) Dijkstra's Shortest Path B) Minimum Spanning Tree
 C) Eulerian Path D) Maximal Flow
10. What is the worst- case time complexity of Prim's algorithm if the adjacency matrix is used? 1 2 3 2
 A) $(\log V)$ B) $O(E^2)$
 C) $\Omega(V^2)$ D) $O(V \log E)$
11. If the keys of Optimal binary Search tree are {10,20,30}, then how many number of trees are possible? 1 3 3 2
 A) 3 B) 7
 C) 5 D) 4
12. Consider a chain of matrices with dimension 2×3 , 3×6 , and 6×4 . What is the minimum number of scalar multiplications needed to multiply these three matrices together? 1 2 3 2
 A) 72 B) 96
 C) 144 D) 64
13. How many solutions exists for placing 4 queens on a 4×4 chessboard so that no two queens threaten each other? 1 2 4 2
 A) 1 B) 2
 C) 4 D) 5
14. Choose the correct statement from the following. 1 1 4 2
 A) branch and bound is more efficient than backtracking B) branch and bound is not suitable where a greedy algorithm is not applicable
 C) branch and bound divides a problem into at least 2 new restricted sub problems D) backtracking divides a problem into at least 2 new restricted sub problems
15. If a data analyst is working with a graph to model website traffic, involving various paths through which users can navigate from one page to another, and needs to analyze potential improvements to user navigation, which algorithm would provide the most comprehensive insight? 1 2 4 2
 A) Floyd-Warshall Algorithm B) Dijkstra's Algorithm
 C) BFS Algorithm D) DFS Algorithm
16. Breadth first search is started on a binary tree beginning from a root vertex. There is a vertex at a distance 4 from the root. If t is the n -th vertex in this BFS traversal, then the maximum possible value of n is ----- 1 2 4 2
 A) 15 B) 31
 C) 7 D) 63
17. Consider a simple graph G with 18 vertices. What will be the size of the maximum independent set of G , if the size of the minimum vertex cover of G is 10? 1 2 5 2
 A) 8 B) 18
 C) 28 D) 10

18. During a series of interviews, a company utilizes a strategy that combines both random selection and ranking to decide on hires. This is to minimize the total hiring cost. What concept from algorithm design does this most closely resemble? 1 2 5 2
 A) Greedy algorithm B) Dynamic programming
 C) Las Vegas algorithm D) Monte Carlo algorithm
19. Problems that can be solved in polynomial time are known as ----- 1 1 5 2
 A) Decision B) Complete
 C) Tractable D) Intractable
20. Which of the following is the application of Maximum Sum Sub-Array problem? 1 1 5 1
 A) Image Processing B) Cryptography
 C) Computer Networks D) Cyber Security

PART - B (5 x 8 = 40 Marks)

Marks BL CO PO

Answer ALL Questions

- 21 a. Illustrate Big oh Notation, Omega Notation and Theta Notations. Depict the same graphically and explain. 8 1 1 1
 (OR)
 b. Find the time complexity of the below recurrence relation 8 1 1 1
 i) $T(n) = \begin{cases} 2T(n-1) & \text{if } n > 0 \\ 1 & \text{otherwise} \end{cases}$
 ii) $T(n) = \begin{cases} 2T(n/2)+1 & \text{if } n > 1 \\ 1 & \text{otherwise} \end{cases}$
- 22 a. i) Solve using substitution method. 8 2 2 1
 $T(n) = T(n-1) + 3; t(1) = 4.$
 ii) Compare and contrast Merge sort and Quick sort.
 (OR)
 b. Illustrate Strassen matrix multiplication. Explain how Strassen Matrix Multiplication algorithms reduces time complexity of matrix multiplication. Write the recurrence relation of Strassen Matrix Multiplication and solve it to find its time complexity. 8 2 2 4
- 23 a. Suppose you are a software engineer working for a digital library that needs to efficiently store and transmit text documents. One of the key challenges is to minimize the amount of storage space required while ensuring fast access to the documents. To address this challenge, you are decided to implement a greedy algorithm. To start with Your client has provided a large document containing the minutes of important meetings, which needs to be stored efficiently. The document consists of 100k characters, with varying frequencies say 'P': 45k, 'Q': 13k, 'R': 12k, 'S': 16k, 'T': 9k, 'U': 5k times of each character. Develop a suitable algorithm to find a solution to reduce storage capacity and analyse its time complexity. 8 3 3 2
 (OR)
 b. Illustrate the concept of Optimal Binary Search tree using dynamic programming with suitable example. 8 3 3 2
- 24 a. In the kingdom of Chessland, the royal gardener is tasked with designing a new garden for the princess's palace. The garden is in the shape of a M x M square, where each square represents a potential location for a flower bed. However, due to the royal decree, no two flowers should be planted in the same row, column, or diagonal. Your task as the royal gardener's assistant is to determine how many different ways there are to plant M flowers in the garden such that no two flowers are in the same row, column, or diagonal. Write an algorithm to calculate the number of valid arrangements for M = 8, and provide the solution with its algorithm using backtracking designing strategy and analysis its time complexity. 8 3 4 2

(OR)

- b. Imagine you're a logistics manager for a distribution company responsible for planning delivery routes for your fleet of trucks. Your company operates in a city with multiple warehouses and delivery locations. Your goal is to minimize the total distance traveled by your trucks while ensuring that each delivery location is visited exactly once. Using the branch and bound algorithm, outline the steps you would take to find the optimal delivery route for your trucks. Consider factors such as the number of delivery locations, the distances between them, and how you would efficiently explore the solution space to find the best route.

8 3 4 2

	1	2	3	4	5
1	∞	20	30	10	11
2	15	∞	30	10	11
3	3	5	∞	2	4
4	19	6	18	∞	3
5	16	4	7	16	∞

- 25 a. Explain about NP-Hard and NP-Complete problems.

8 2 5 2

(OR)

- b. Construct an algorithm for randomized quick sort and perform the complexity analysis of randomized quick sort algorithm.

8 2 5 2

PART - C (1 x 15 = 15 Marks)

Marks BL CO PO

Answer ANY ONE Question

26. Construct an algorithm for Fractional knapsack problem. Illustrate the same for the given set of items and knapsack capacity = 60 kg. Find the optimal solution for the fractional knapsack problem making use of greedy approach.

15 3 3 2

Item	Weight	Value
1	5	30
2	10	40
3	15	45
4	22	77
5	25	90

27. Explain Vertex Cover Problem briefly. Develop a greedy based approach to determine the minimum size vertex cover of a graph with n nodes and m edges of the following graph. Also find its time complexity.

15 2 5 2


